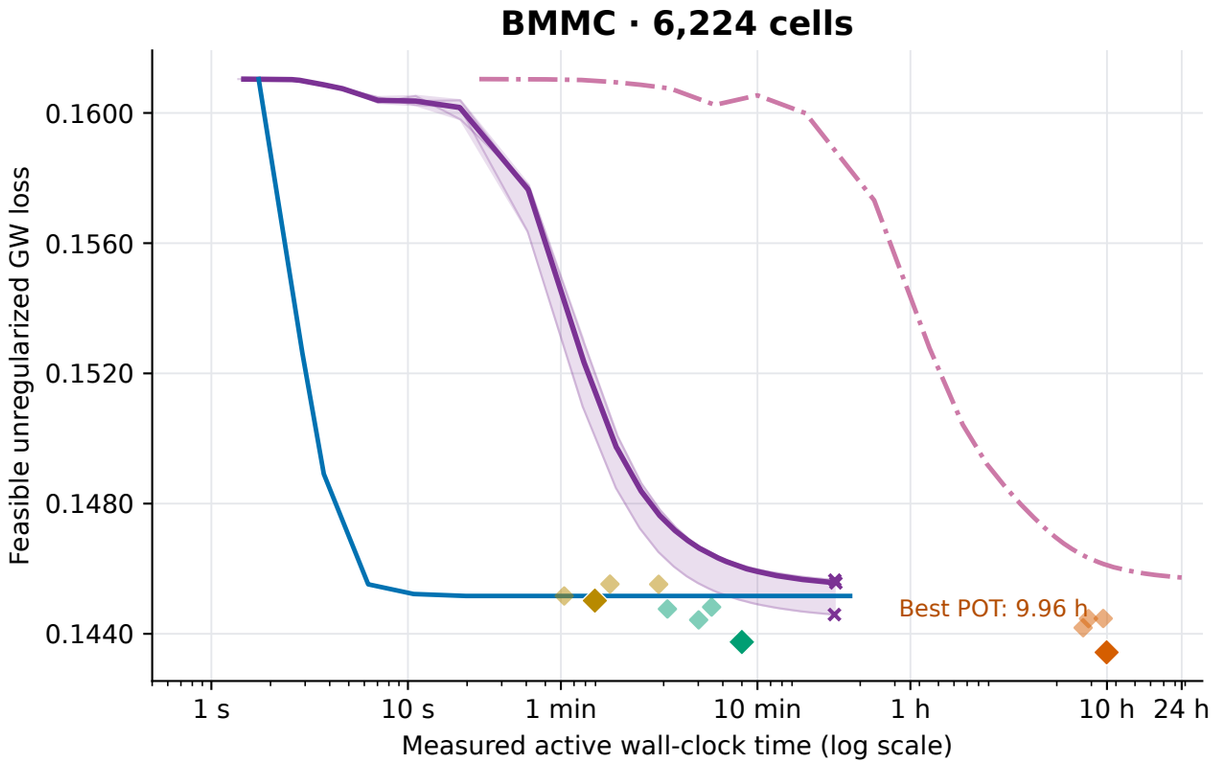
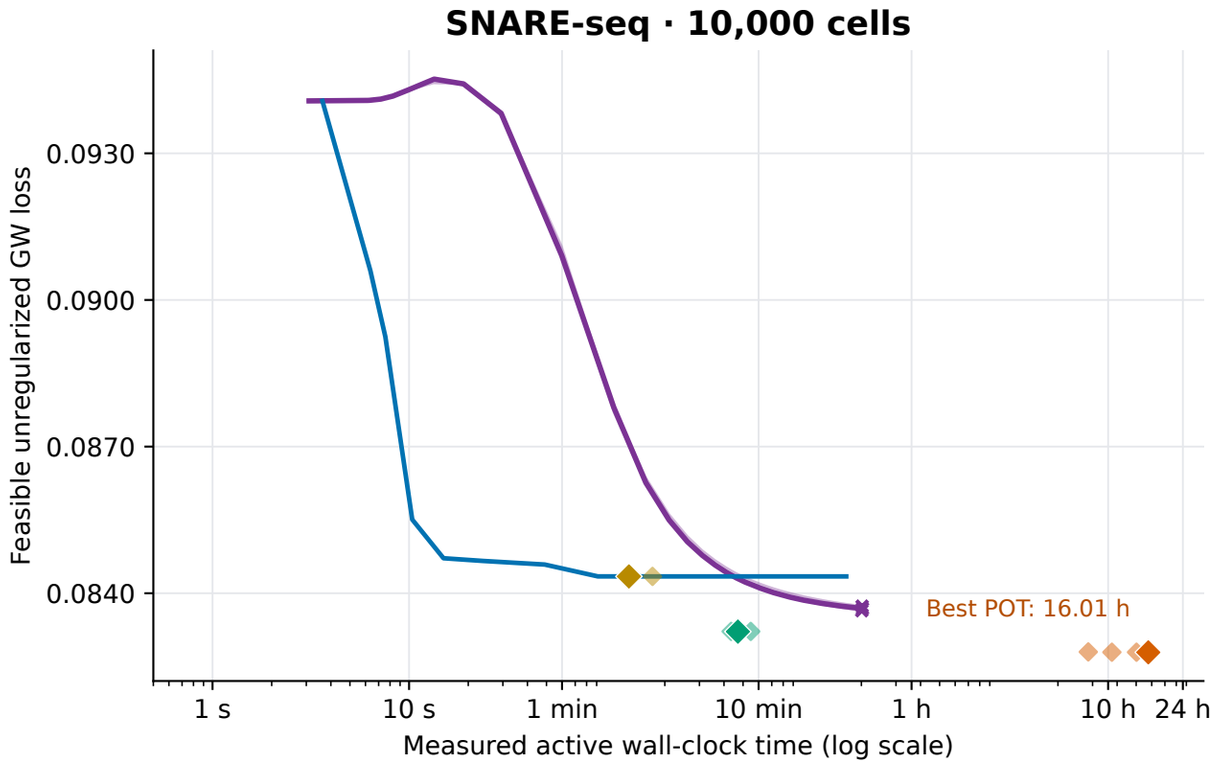
GW loss versus measured wall-clock time

FP64 · One A100 per GPU run; CPU run explicitly marked · Curves use recorded timestamps · Logarithmic time axis

FRAMES, 1,024 pairs, A100: median and range of 3 seeds
Full-gradient POT, ε=10⁻³, product start: 2,000 updates
FRAMES, 4,096 pairs, CPU: seed 811
POT sweep, ε=10⁻³: completed-run endpoints
POT sweep, ε=3×10⁻⁴: completed-run endpoints
POT sweep, ε=10⁻⁴: completed-run endpoints





Lines: timed replay (POT 2,000 updates; FRAMES 20,000). Diamonds: all completed starts in the entropy sweep; larger = lowest loss at that ε.

The sweep saved total runtime but no loss timestamps: diamonds are final results only, not reconstructed trajectories. Unfinished sweep runs omitted here.

Time includes data loading, optimization, loss diagnostics and checkpoint work; excludes queueing, prefactor tuning and runtime initialization.

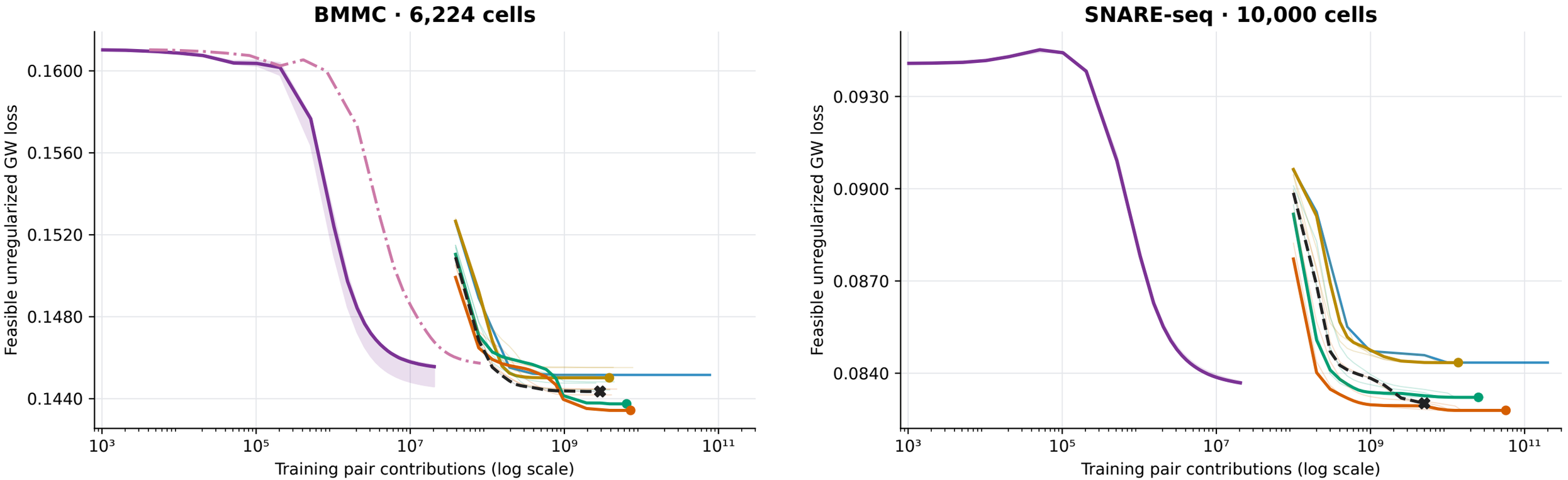
FRAMES time median/range interpolate only within shared observed time support. CPU and A100 timings are different hardware comparisons.

POT sweep starts vary; FRAMES starts from the product coupling. The 2,000-update POT replay has no early stopping; the sweep does.

GW loss versus training oracle work

Pair contributions at accepted updates · FRAMES: actual pair draws · Full-gradient POT: equivalent complete weighted sums

FRAMES, 1,024 pairs, A100: median and range of 3 seeds
Full-gradient POT, ε=10⁻³, product start: 2,000 updates
FRAMES, 4,096 pairs, CPU: seed 811
POT sweep, ε=10⁻³: best completed start
POT sweep, ε=3×10⁻⁴: best completed start
POT sweep, ε=10⁻⁴: best completed start
POT sweep, ε=3×10⁻⁵: unfinished product start



After k accepted updates: FRAMES counts qk sampled pairs; full-gradient POT counts nmk weighted-pair equivalents (n × m coupling).

POT evaluates the full sum using matrix products; nm equivalents are not nm literal stochastic-oracle calls. This axis is not FLOPs or runtime.

Loss diagnostics and interrupted/retried updates are excluded. Sinkhorn iterations reuse the gradient and add no gradient calls.

Lines connect diagnostics; zero-work initialization is omitted. Faint POT lines: other starts. Unfinished endpoints: BMMC k=75; SNARE k=50.

Counting one batched gradient evaluation as one call instead gives k calls for either method, reproducing the iteration axis.